Course Code : ECT 251 **KQLR/RS – 24 / 1543**

Third Semester B. Tech. (Electronics and Communication Engineering) Examination

ELECTRONIC DEVICES

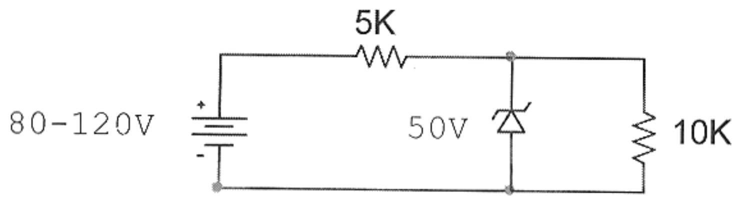
Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) Assume suitable data wherever necessary.

(2) All questions are compulsory.

1. (a) For the circuit shown below, Evaluate the maximum and minimum values of Zener diode current. Also find P_{Zmax}.



5(CO1,5)

(b) A full wave rectifier circuit design with two identical silicon diodes with a forward resistance of 10 Ω each. A d.c. voltmeter connected across the load of 1 KΩ reads 60 V.
Calculate :

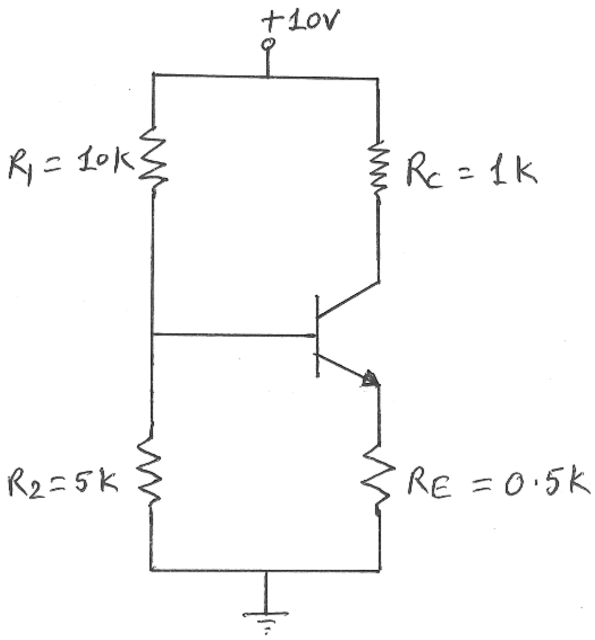
(i) Percentage efficiency.

(ii) Ripple factor of FWR.

(iii) Transformer voltage rating. 5(CO1,5)

2. (a) Draw the input, output and transfer characteristics of transistor operated in Common Emitter configuration. Derive an expression of output collector current. 4(CO1)

(b) Determine the collector current and collector voltage of silicon transistor operated in CE mode with Self bias circuit shown below, Assume $\beta = 100$.



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6(CO4,5)

3. (a) A Common Emitter amplifier with voltage divider bias has $R_1 = 100\ K\Omega$, $R_2 = 10\ K\Omega$, $R_C = 5\ K\Omega$ and $R_S = 10\ K\Omega$. Draw hybrid model for the voltage divider bias. Use following h-parameters and estimate A_I, A_V, A_P, input and output impedances.

$h_{ie} = 1100\ \Omega$, $h_{re} = 2.5 \times 10^{-4}$, $h_{fe} = 50$, $h_{oe} = 25$ uA/V.

8(CO4,5)

(b) Draw h-parameter equivalent model for the common base configuration. 2(CO1)

4. (a) The FET amplifier with $I_{DSS} = \mathbf{15}$ mA, $V_p = \mathbf{-6}$ V. Determine the drain current for $V_{GS} = \mathbf{-3}$ V. 5(CO4,5)

(b) Describe the use of FET as a voltage variable resistance. 5(CO2)

5. (a) Discuss depletion and enhancement type MOSFETs in respect of working and C-V characteristics. Also write their individual applications. 6(CO2)

(b) What do you mean by body effect in MOSFET devices ? 4(CO2)

6. (a) Implement the equation using CMOS logic :

$X = (A + B) (C + D) + E$ 6(CO3)

(b) Draw and discuss voltage transfer characteristics of CMOS invertor. Define V_{IH} and V_{IL}. 4(CO2,3)